



POLITECNICO
MILANO 1863

SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE

In-plane Finite Element Model of a Truss Bridge

ADVANCED DYNAMICS OF MECHANICAL SYSTEMS
LAB ACTIVITY REPORT

Authors: **Simone Falsitta, Francesca Gaudiano,
Gabriele Massa, Alessandro Morini**

Student ID: 10765786, 10813191, 10796656, 10773766

Advisor: Prof. Roberto Corradi

Co-advisor: Prof. Ivano La Paglia

Academic Year: 2024-25

Contents

Contents	i
1 Undeformed Structure	1
1.1 Mesh Definition	1
1.2 Nodes and Elements	1
2 Natural Frequencies and Mode Shapes	3
2.1 Natural Frequencies	3
2.2 Mode Shapes	3
3 Frequency Response Functions	7
3.1 Introduction of Damping	7
3.2 Computation of FRF	7
3.3 Internal Actions	9
3.4 Constraint Forces	11
4 Modal Superposition Approach	13
5 TMD Design	15
5.1 Defining the TMD's parameters	15
5.2 Installation of the TMD	17
6 Static Response	19
6.1 Deck Weight	19
6.2 Weight of the Entire Structure	20
7 Critical Velocities	21

1 | Undeformed Structure

1.1. Mesh Definition

The first thing to introduce when approaching a Finite Element model is the **mesh** used for the discretization of the continuous rigid body analyzed. In this study-case, the FEM analysis will be conducted using Euler-Bernoulli **beam elements**. Before the discretization, the first check to carry out is about the maximum length of each element, that has to be set so that every element always work in **quasi-static conditions** inside the assigned frequency range (**0-7 Hz** in this case).

The computation of the maximum length is performed considering a pinned-pinned element, for the simplicity of its boundary conditions. We impose, for the generic ' k^{th} ' element, that:

$$\omega_k^{(1)} = 1.5 \cdot \Omega_{max}$$

From this, knowing the analytical expression for the first natural frequency for a pinned-pinned beam, it is possible to derive the maximum length L_{max} of the elements:

$$L_{max} = \sqrt{\frac{\pi^2}{1.5 \cdot \Omega_{max} \cdot \sqrt{(\frac{EJ}{m})_{min}}}}$$

Knowing the different properties in terms of mass and EJ of the different beams that are composing the bridge, we end up with:

$$L_{max} = 7.89m$$

1.2. Nodes and Elements

Knowing the maximum length of each element it is now possible to properly discretize the in-plane bridge structure, compiling the **INPUT FILE**, in which are reported all the pieces of information regarding the nodes, their position and numbering, the elements and their mass properties. Running the input file into a dedicated *Matlab* code, the assembling

of the mass matrix $[M]$ and stiffness matrix $[K]$ is performed, and it is also possible to check the result of the discretization from a graphical point of view. The analyzed bridge in the FEM model will be represented with **36 nodes** and **46 elements**, with **9 different material properties**, as shown in Figure 1.1.

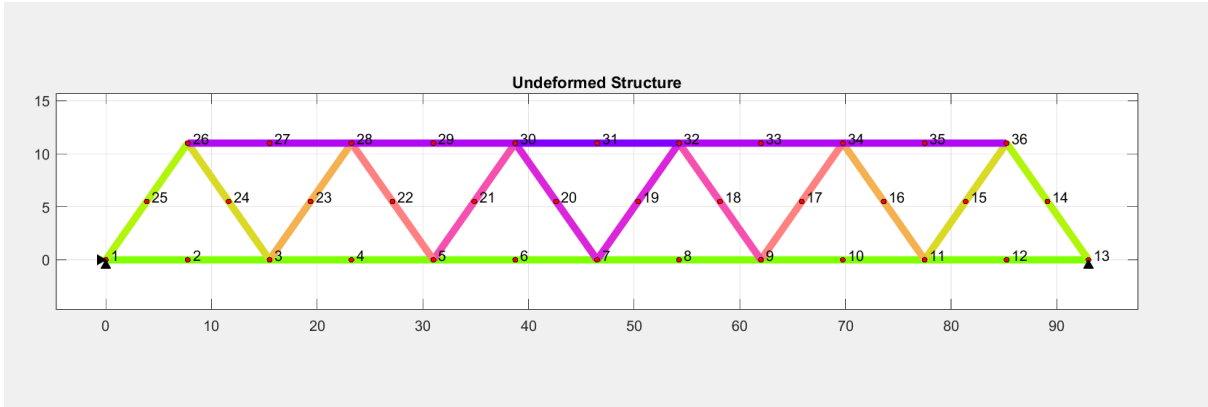


Figure 1.1: Undeformed Structure



Figure 1.2: Joncherolles Bridge, France, 1930-40



Figure 1.3: Real truss bridge under construction

2 | Natural Frequencies and Mode Shapes

2.1. Natural Frequencies

Since the overall Mass and Stiffness matrices are known, it is possible to introduce a partition that accounts for Free and Constrained degrees of freedom. Starting from the motion equation of the Free-Free subsection it's possible to compute the natural frequencies, as:

$$[M_{FF}]\ddot{\underline{x}}_{FF} + [K_{FF}]\underline{x}_{FF} = 0$$

$$(-\omega^2[I] - [M_{FF}]^{-1}[K_{FF}])\underline{X} = 0$$

The natural frequencies ω are the **eigenvalues** of $([M_{FF}]^{-1}[K_{FF}])$, while the modes shapes are the **eigenvectors**. In Table 2.1 the values for the first six natural frequencies of the bridge are shown.

Nat. Freq	Value [Hz]
ω_1	2.52
ω_2	5.24
ω_3	5.56
ω_4	5.67
ω_5	6.40
ω_6	6.87

Table 2.1: Natural Frequencies

2.2. Mode Shapes

In the following figures are represented the modes shapes of the bridge, up to the sixth one. The modes are plotted assuming a scaling factor of 10.

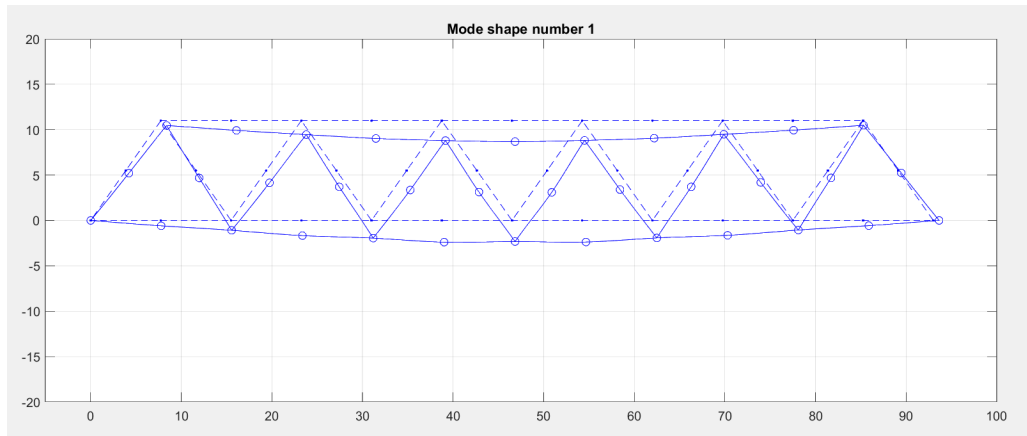


Figure 2.1: First mode: 2.52 Hz

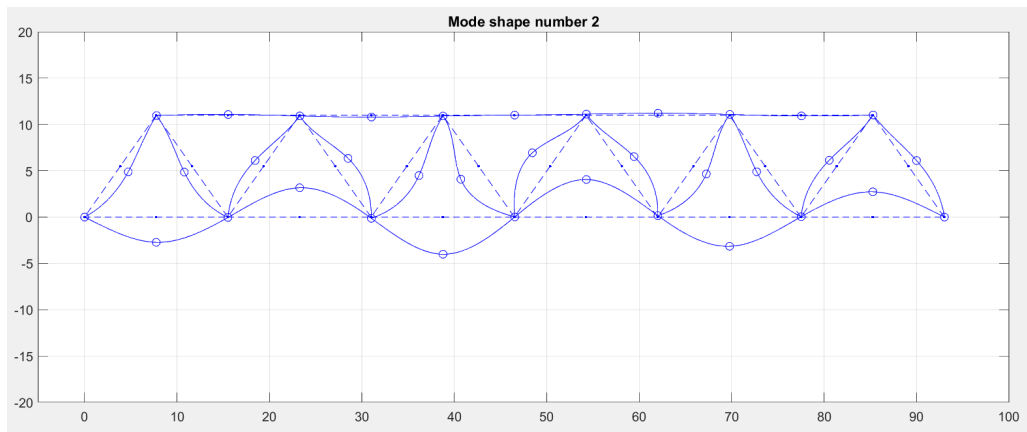


Figure 2.2: Second mode: 5.24 Hz

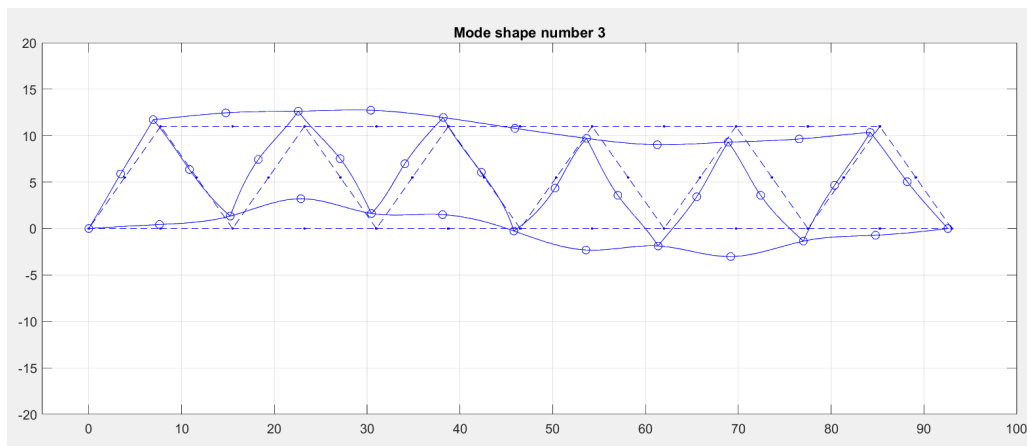


Figure 2.3: Third mode: 5.56 Hz

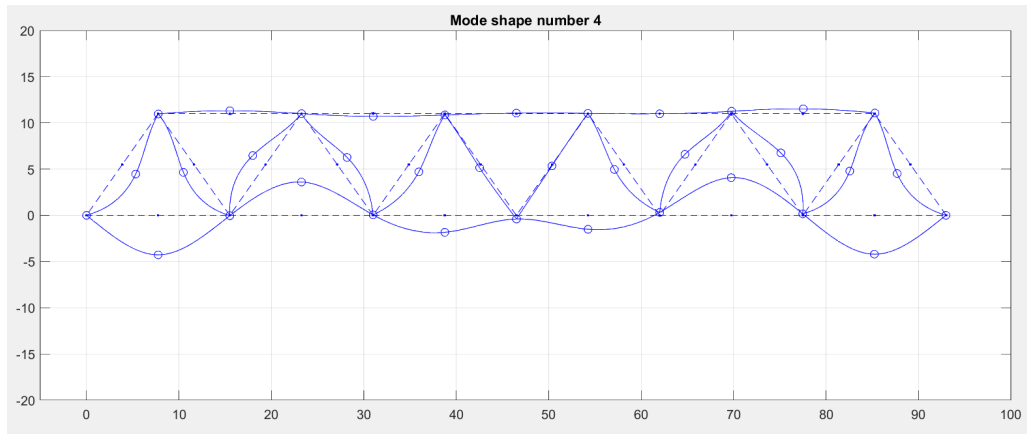


Figure 2.4: Fourth mode: 5.65 Hz

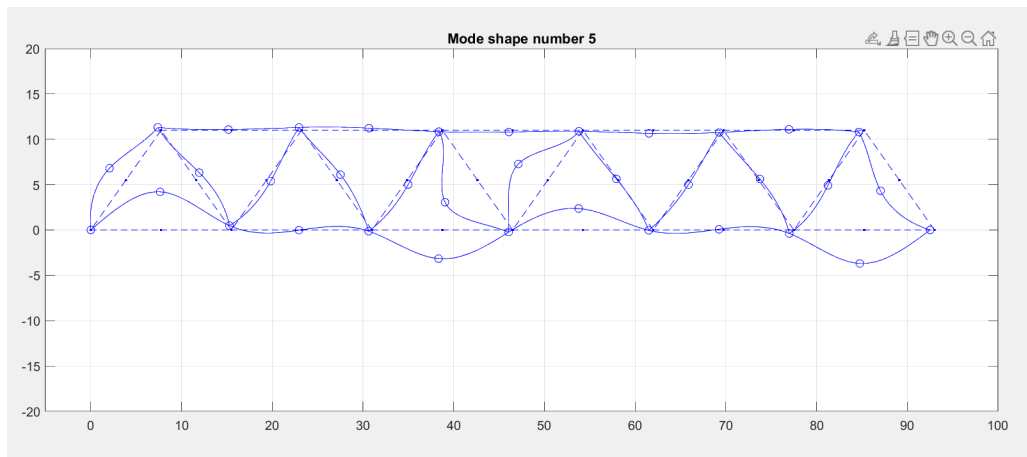


Figure 2.5: Fifth mode: 6.40 Hz

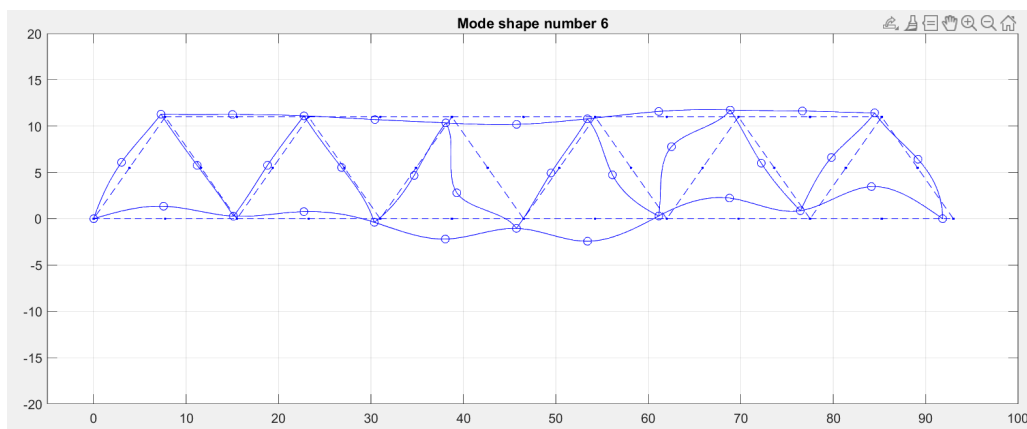


Figure 2.6: Sixth mode: 6.87 Hz

3 | Frequency Response Functions

3.1. Introduction of Damping

Damping is introduced in the equation of motion thanks to the **Rayleigh model**:

$$[C_{FF}] = \alpha[M_{FF}] + \beta[K_{FF}]$$

The values for α and β are computed basing on experimental measurements. In this example, the values are chosen in order to obtain $\xi_1 = 0.01$ for the first mode and $\xi_2 = 0.0075$ for the second mode. Relying on the relation:

$$\xi_i = \frac{\alpha}{2\omega_i} + \frac{\beta\omega_i}{2}$$

the values for α and β are obtained and the matrix $[C_{FF}]$ is introduced.

3.2. Computation of FRF

The Frequency Response Functions show the dynamic behavior of the bridge (in terms of magnitude and phase) when subjected to an external load. In the case considered the impulse force is applied on node 6 (Point A), directed upwards. The FRFs in the following pictures refer to the response, both in terms of displacement and acceleration (**receptance and inertance**) of Point A (co-located FRF), Point B (mid-span of the deck, node 7) and Point C (node 31). For the plot, the matrix $[idb]$ has been used for the extraction of the correct single FRF from the Frequency Response Matrix $[X]$, that accounts for the response to the input of all the nodes, in the three direction (x, y, θ).

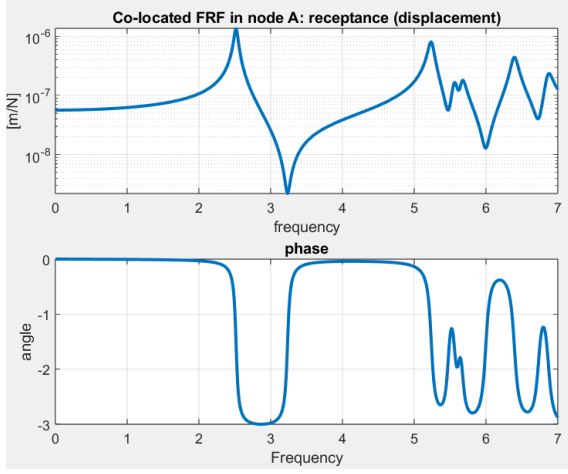


Figure 3.1: FRF in A: receptance

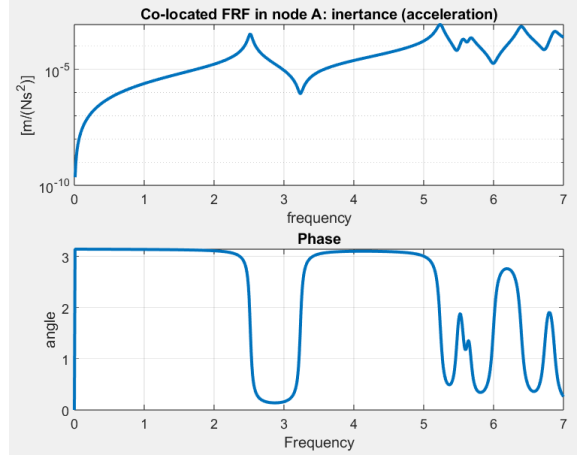


Figure 3.2: FRF in A: inertance

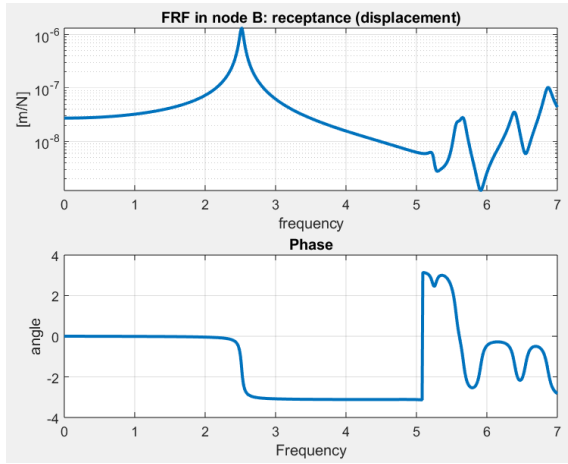


Figure 3.3: FRF in B: receptance

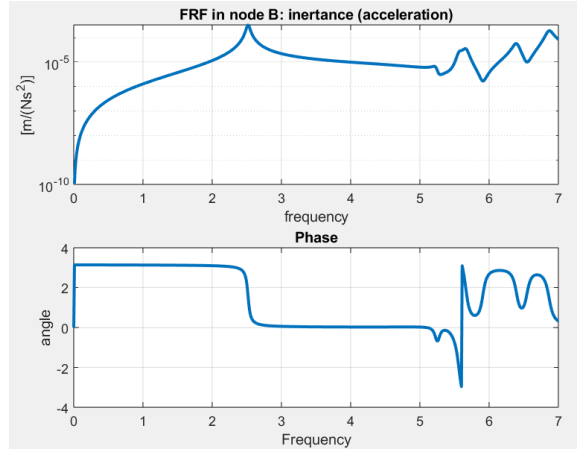


Figure 3.4: FRF in B: inertance

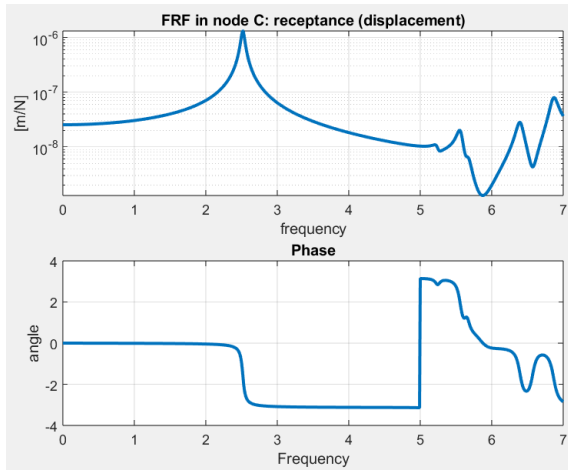


Figure 3.5: FRF in C: receptance

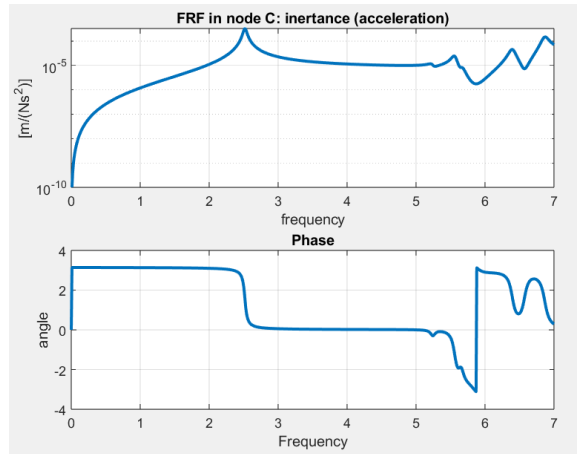


Figure 3.6: FRF in C: inertance

3.3. Internal Actions

The internal actions in node D and C are computed in function of the nodal displacements by means of the **shape functions**. Assuming the shape function $u(\xi, t)$ for the axial displacement and $w(\xi, t)$ for the transverse one, it is possible to refer to the internal actions N, T, M:

$$N = EA \frac{\partial u}{\partial \xi}$$

$$T = EJ \frac{\partial^3 w}{\partial \xi^3}$$

$$M = EJ \frac{\partial^2 w}{\partial \xi^2}$$

Since u is a linear function of ξ and w is a cubic function of ξ , we expect the axial force and the shear force to be constant inside a single element. Since the moment depends on ξ , we consider the FRFs seen by two consecutive elements: the first with the node at the end ($\xi = L_{el}$) and the other with the node of interest at the beginning ($\xi = 0$).

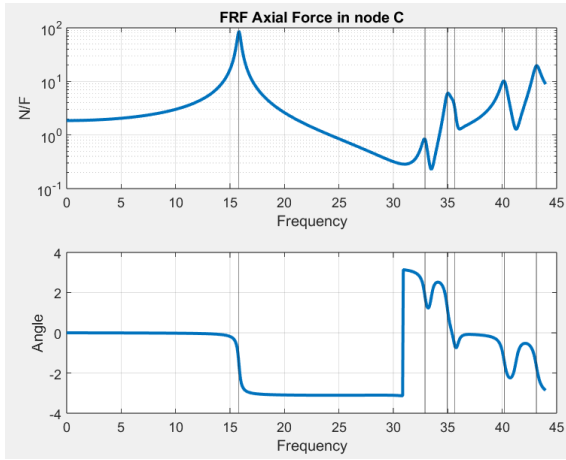


Figure 3.7: Axial FRF in point C

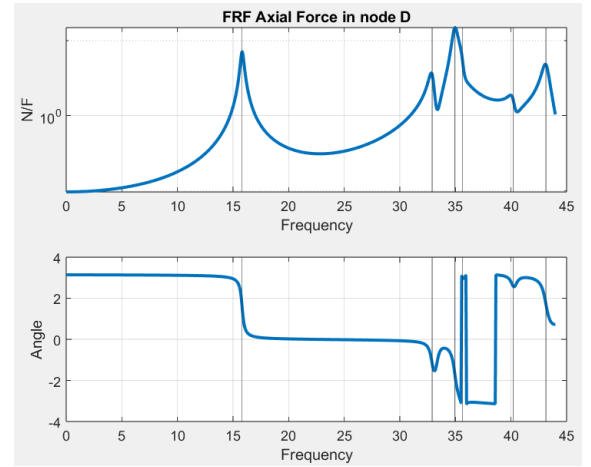


Figure 3.8: Axial FRF in point D

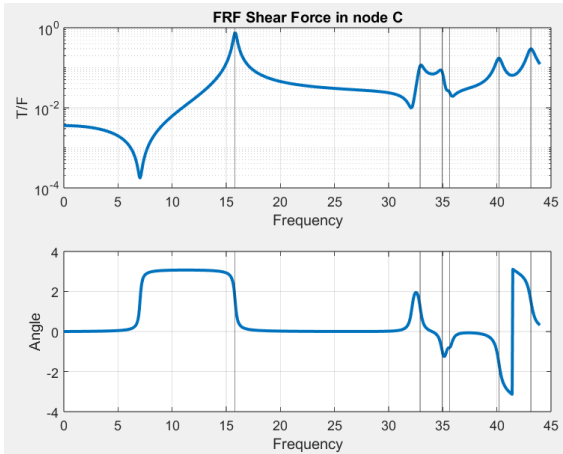


Figure 3.9: Shear FRF in point C

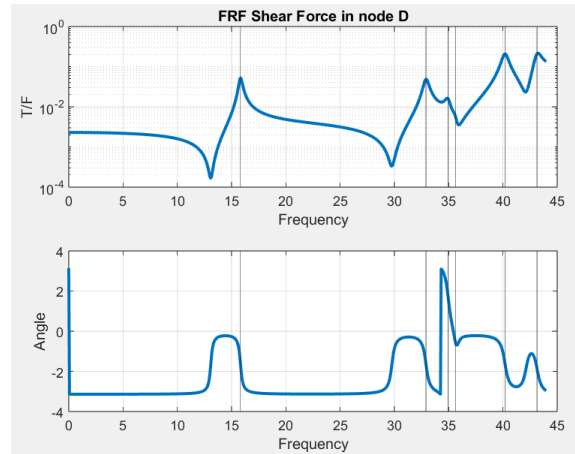


Figure 3.10: Shear FRF in point D

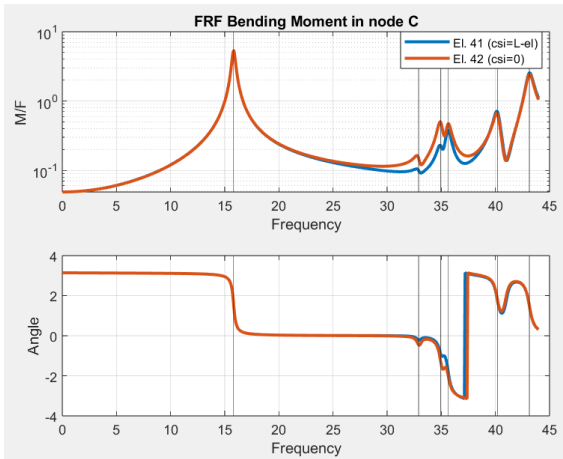


Figure 3.11: Bending Moment FRF in C

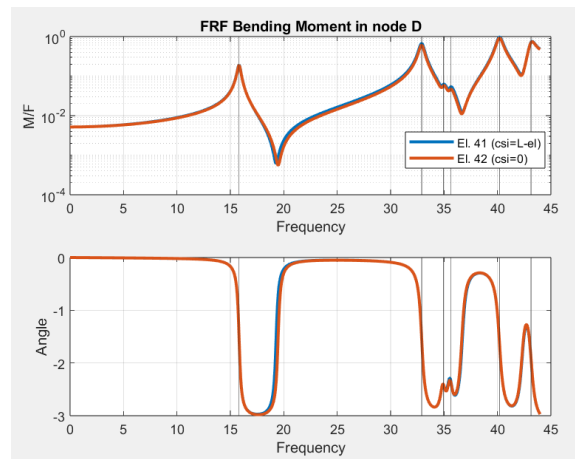


Figure 3.12: Bending Moment FRF in D

3.4. Constraint Forces

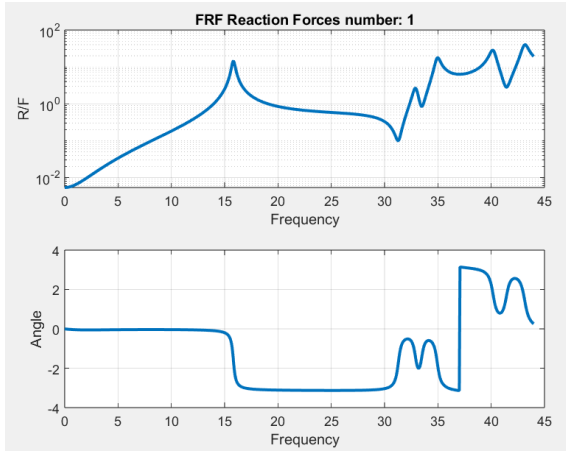


Figure 3.13: Left side, x direction

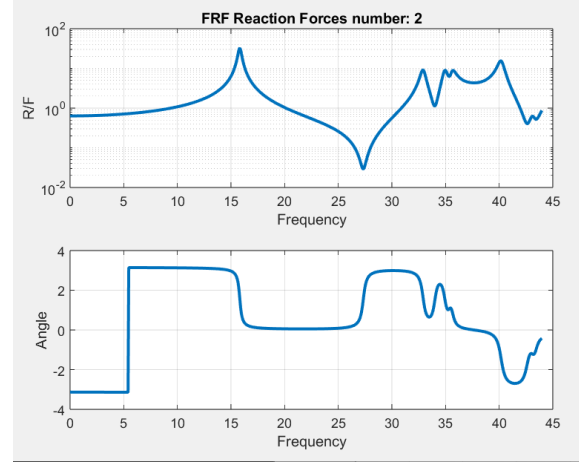


Figure 3.14: Left side, y direction

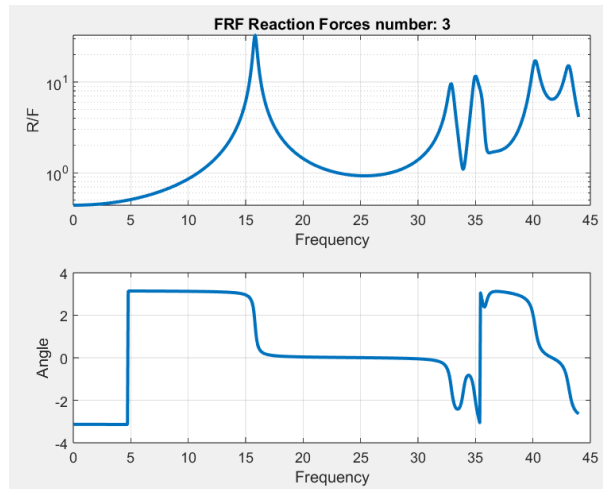


Figure 3.15: Right side, y direction

The FRFs that relate the constraints forces to the input force in node A are built considering the partition Constrained-Free and Constrained-Constrained of the global matrices $[M]$, $[C]$ and $[K]$. Solving the equation in the $\underline{x}_F$ unknown (basing on the F-F and F-C partition) it is possible to compute the reaction forces $\underline{R}$.

4 | Modal Superposition Approach

The aim of this Chapter is to compute the Frequency Response Functions of the structure relying on the modal superposition approach. In particular, we want to calculate the response accounting only for the first three modes. To do that, we define a matrix $[\Phi]$ that allows to move in modal coordinates, defining, as it follows, the Modal Mass, Modal Damping and Modal Stiffness matrices.

$$[\Phi] = [\underline{X}^{(1)} \underline{X}^{(2)} \underline{X}^{(3)}]$$

$$[M_{mod}] = [\Phi]^T [M] [\Phi]$$

$$[C_{mod}] = [\Phi]^T [C] [\Phi]$$

$$[K_{mod}] = [\Phi]^T [K] [\Phi]$$

$$[Q_{mod}] = [\Phi]^T \underline{F}$$

It's possible to plot the FRFs obtained with the Modal Approach comparing the graphs with the results obtained in Section 3.2: we note that especially for the third resonant peak the results given by the modal approach are quite far from the complete FRF. This behavior may be caused by the facts that we have many resonances in a narrow frequency range.

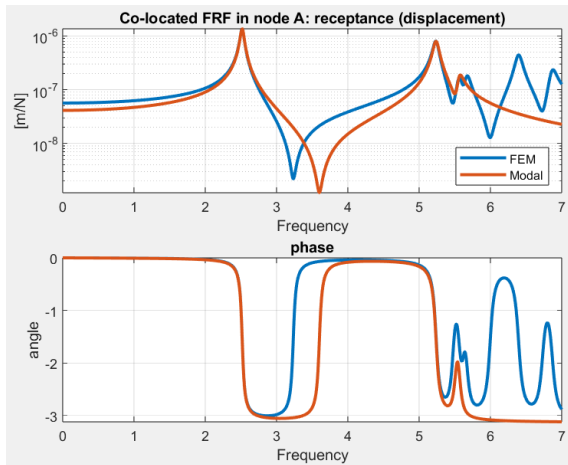


Figure 4.1: FRF in A: receptance

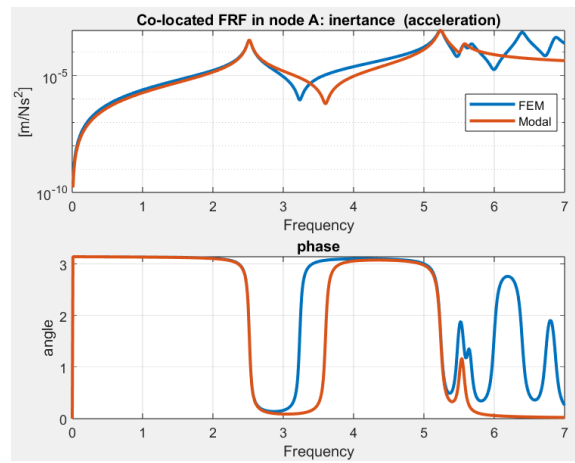


Figure 4.2: FRF in A: inertance

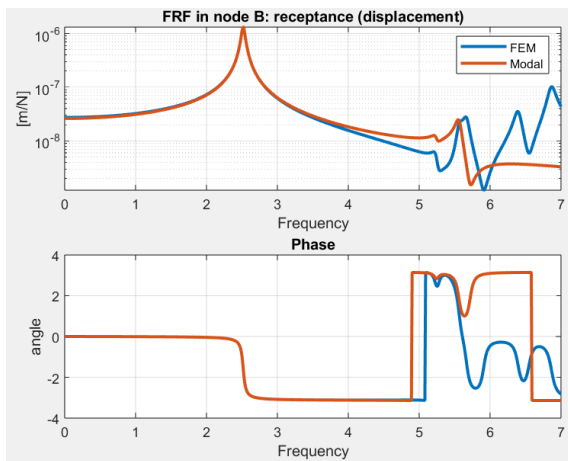


Figure 4.3: FRF in B: receptance

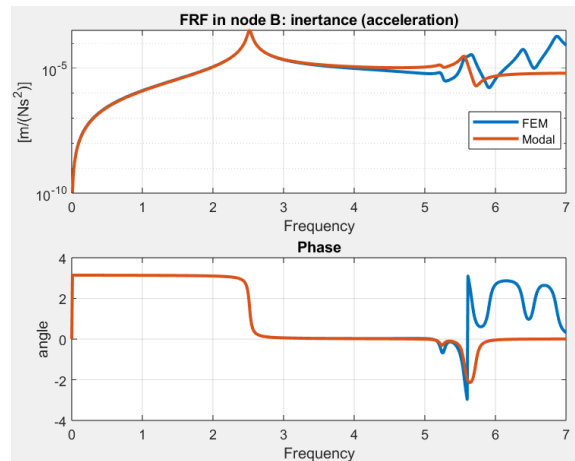


Figure 4.4: FRF in B: inertance

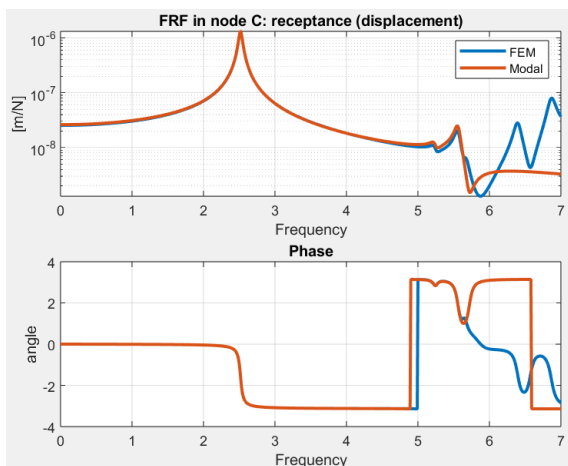


Figure 4.5: FRF in C: receptance

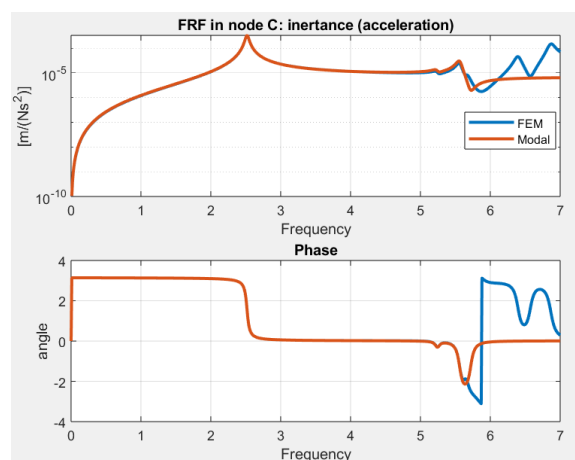


Figure 4.6: FRF in C: inertance

5 | TMD Design

5.1. Defining the TMD's parameters

We want to design and install over the bridge a Tuned Mass damper (TMD) in order to damp out (at least of 15%) the resonant peak associated with the first mode. In order to properly choose the mass, the stiffness and the damping of the TMD that we are going to design, it's useful to rely on the **Modal Approach**, considering a reduction for the first only mode. In that way the whole bridge is behaving like a 1-dof **harmonic oscillator**, in which the characteristic parameters are the modal mass, the modal damping and the modal stiffness. Writing the equation of motion of the 2 degrees-of-freedom system formed by the "reduced" bridge and the TMD it is possible to evaluate which combination of parameters for the TMD best suits our goal. Proceeding with successive approximations we found the best values for μ , the mass ratio and ξ , the adimensional damping ratio. In particular:

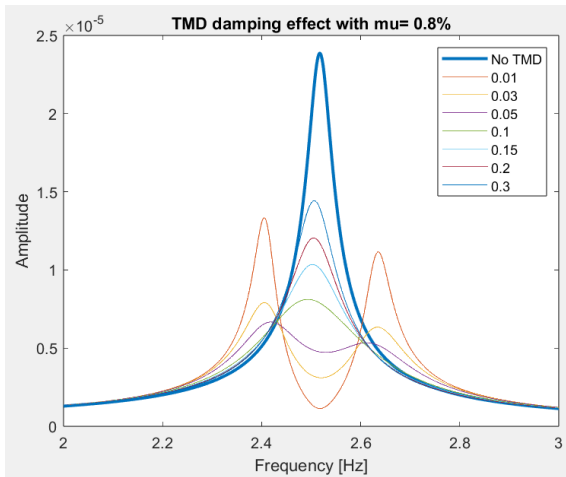


Figure 5.1: Case: $\mu = 0.8\%$

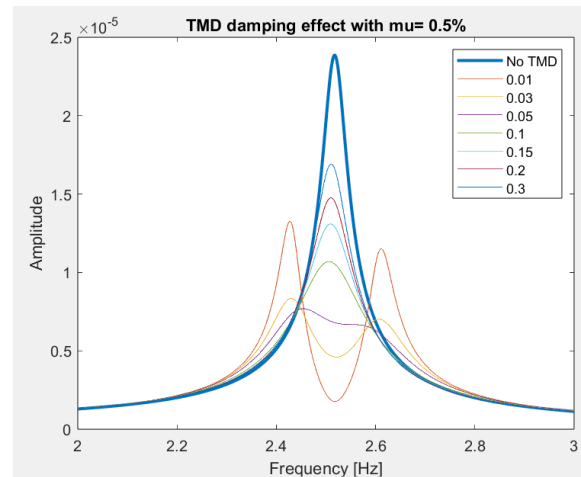
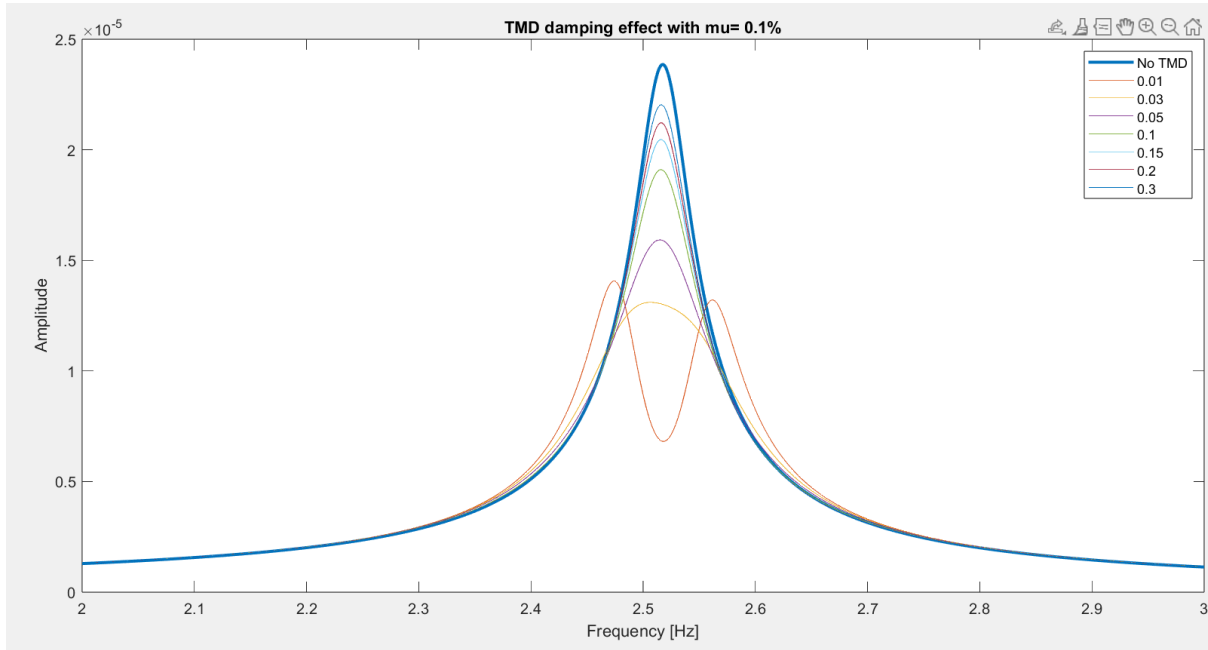


Figure 5.2: Case: $\mu = 0.5\%$

Figure 5.3: $\mu = 0.1\%$

ξ	Damping
0.01	44.6%
0.03	45.1%
0.05	33.2%
0.1	19.9%
0.15	14.2%
0.2	11.0%
0.3	7.6%

Table 5.1: Percentage damping $\mu = 0.1\%$

In order to ensure a proper damping of the peak, installing a limited mass and avoid the formation of two lateral peaks it has been decided to install a TMD having $\mu = 0.1\%$ and $\xi = 0.05$. As a consequence, the real TMD on the bridge will have a mass of $300kg$, and will provide a damping action in the order of 30% to the bridge.

5.2. Installation of the TMD

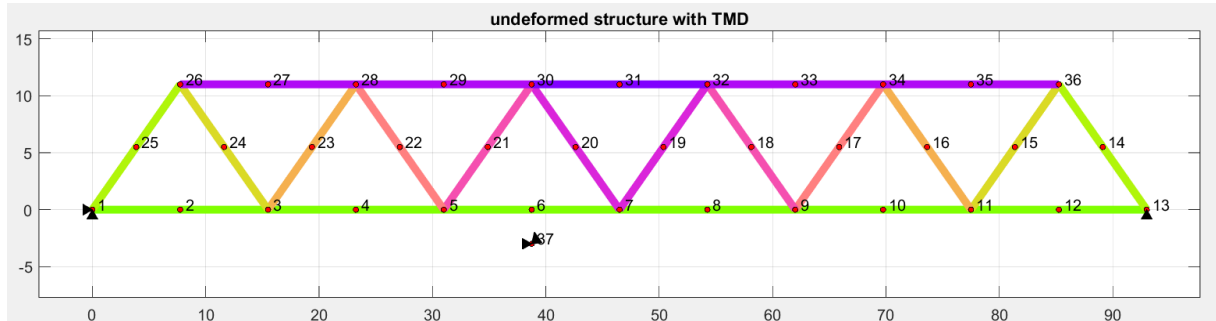


Figure 5.4: TMD positioning: node 6

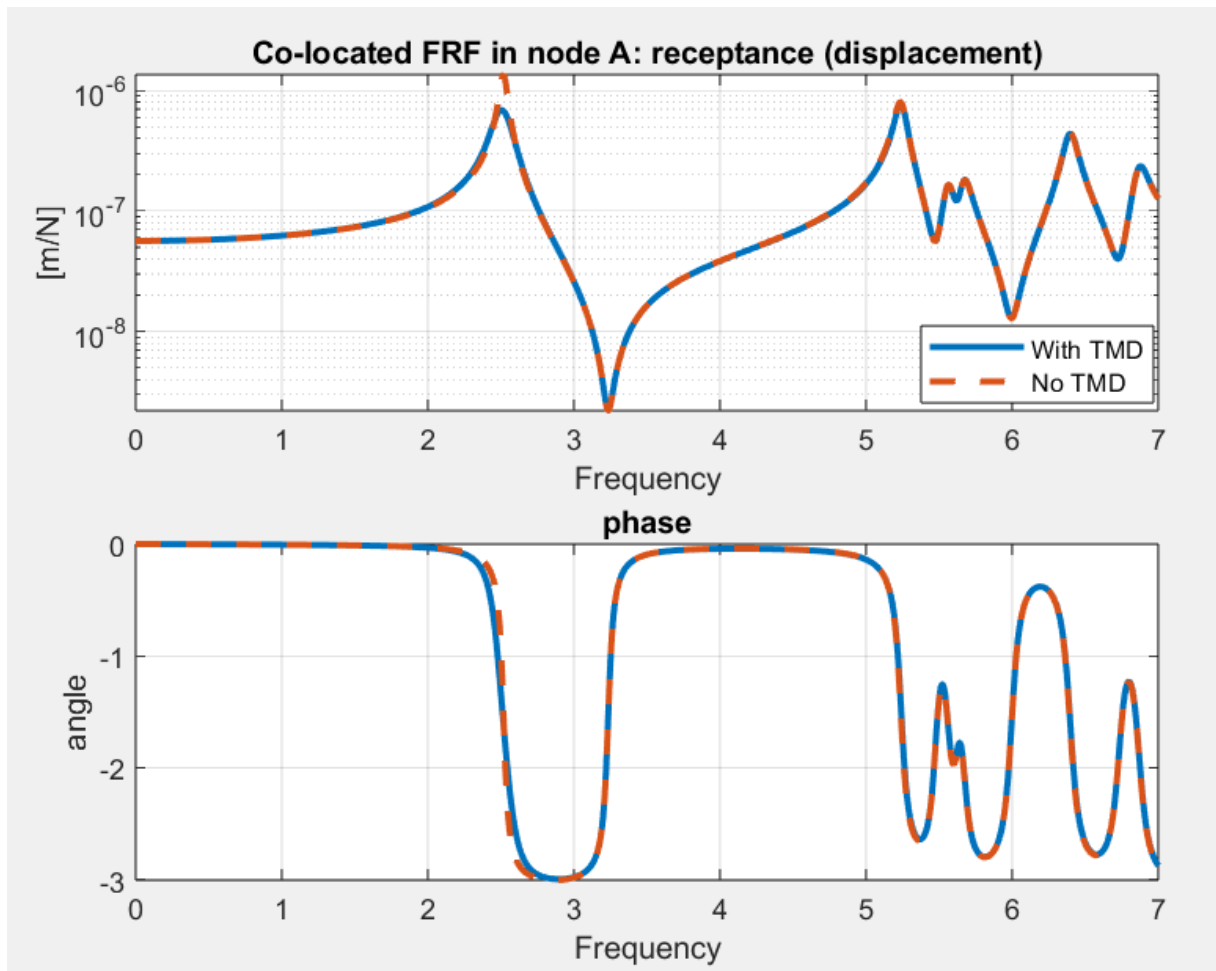


Figure 5.5: TMD: comparison of FRFs

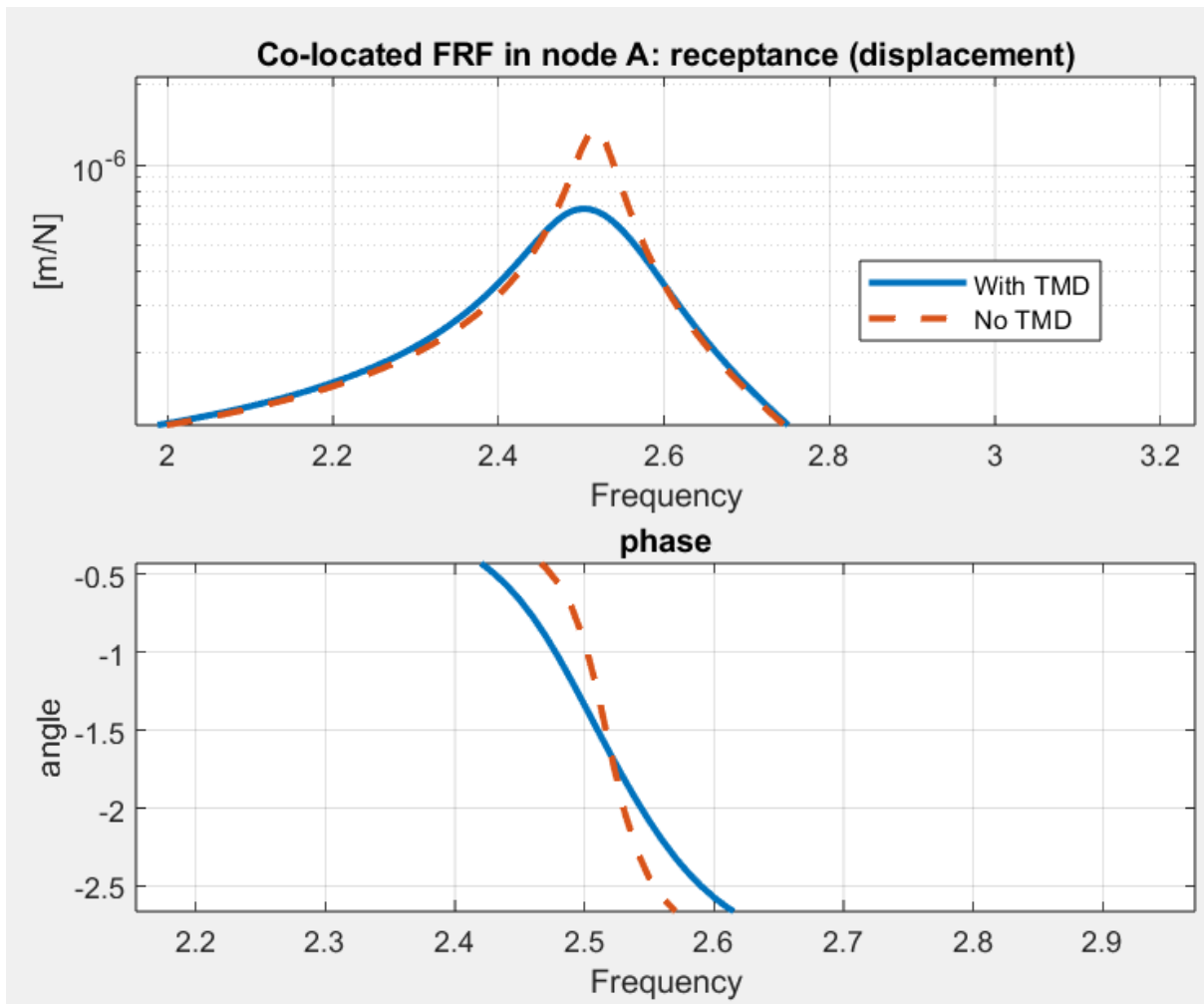


Figure 5.6: Zoom of the first peak

6 | Static Response

For the computation of the static response of the structure under its own weight, two different approaches can be followed. As a first step, the static deformation has been drew considering a **distributed load** over each element, having magnitude equal to the mass per unit length of the element (kg/m) multiplied by the gravity acceleration g . As an alternative, we can directly extract from the **mass matrix** $[M]$ the values of mass that are related to the vertical degree of freedom of each node, and storing them in a proper way we build the vector of the nodal equivalent forces due to the weight. Both methods have been implemented, and the results are here compared.

6.1. Deck Weight

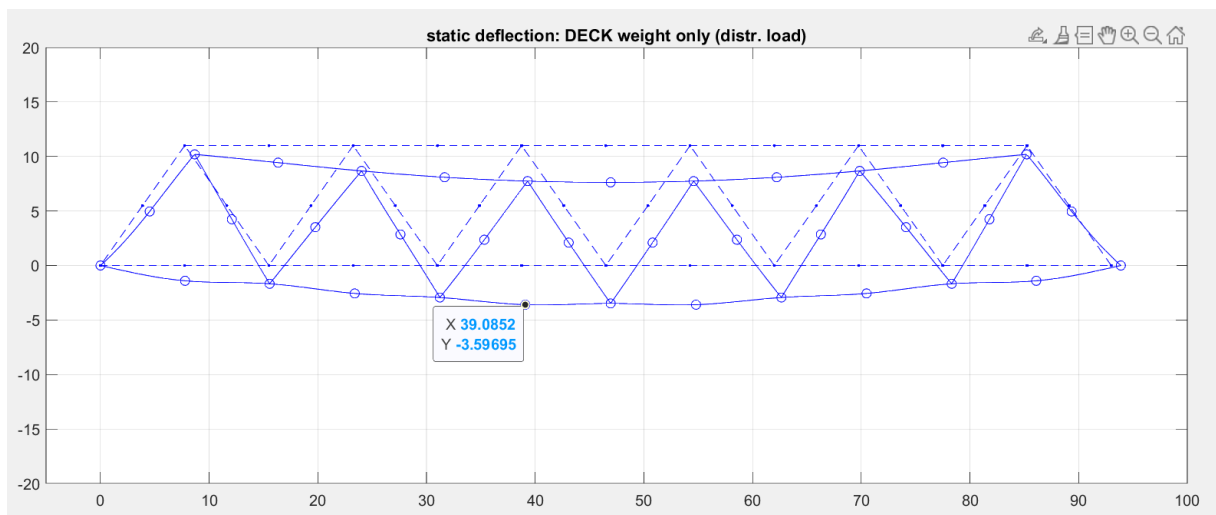


Figure 6.1: DISTRIBUTED LOADS: Scaling factor: 100. Max def.: $-35.9mm$

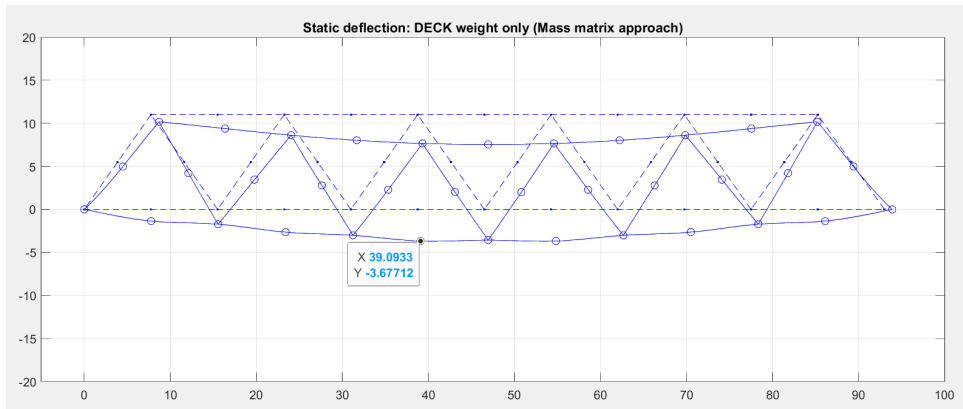


Figure 6.2: MATRIX APPROACH: Scaling factor: 100. Max def.: $-36.8mm$

6.2. Weight of the Entire Structure

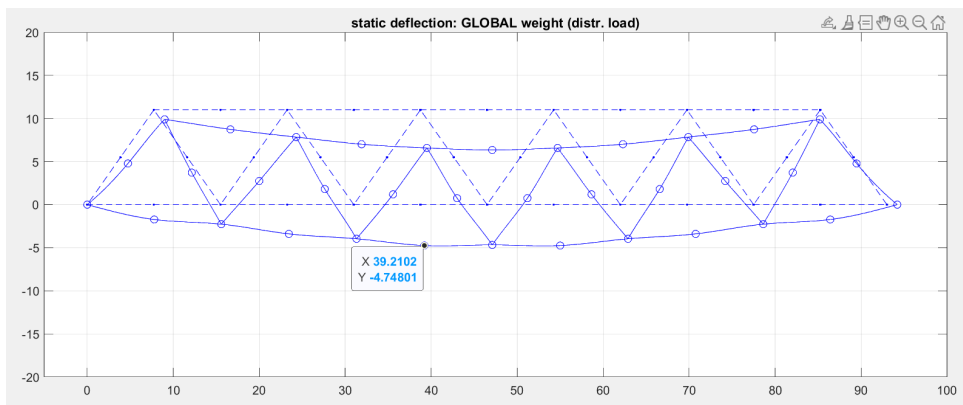


Figure 6.3: DISTRIBUTED LOADS: Scaling factor: 100. Max def.: $-47.5mm$

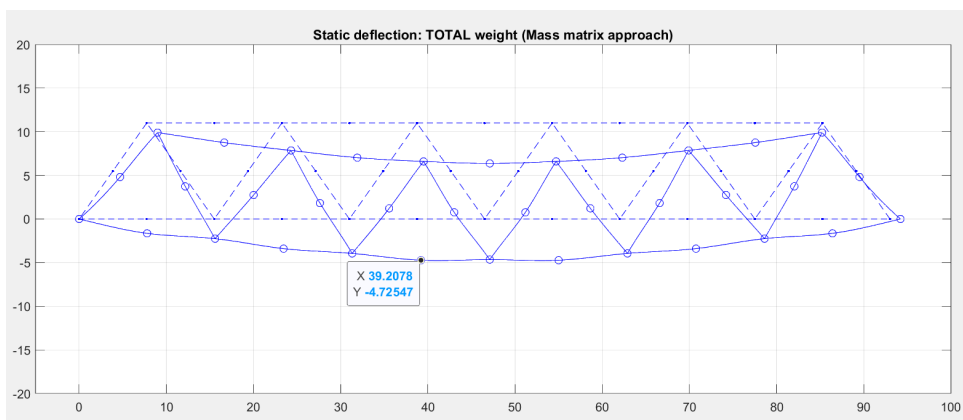


Figure 6.4: MATRIX APPROACH: Scaling factor: 100. Max def.: $-47.7mm$

7 | Critical Velocities

Let's assume that the truss bridge is now runned by a train. The passage of a long vehicle, with many and equally-spaced wheels introduces a periodic forcing on the structure, that can lead to a **resonant excitation**. In particular, if the train is sufficiently long, we can consider a steady state situation, even if, in principle, every single load represent alone a transient excitation (single moving load). Relying on the superposition principle, we can split the frequency of the load in a sum of integer multiple of the fundamental frequency Ω_0 , computed as:

$$\Omega_0 = \frac{2\pi}{\tau} = \frac{2\pi V}{d}$$

where d is the **distance** between the pulses, V is the **speed** of the loads and $\tau = d/V$ is called **periodicity** of the load.

Whenever one of the input frequencies $K \cdot \Omega_0$ **gets tuned** on one of the natural frequencies ω_i of the bridge, the structure can be excited in resonance. This happens when:

$$K\Omega_0 = \omega_i = 2\pi f_i$$

$$K \cdot \frac{2\pi V}{d} = 2\pi f_i$$

$$V = \frac{f_i d}{K}$$

where V is the **critical velocity** that induces resonance. Since, in theory, there are infinite possible couples (K, i) that can be used in the relation, we concentrate our attention on the first 2 modes ($i = 1, 2$) and on the first harmonics of the load sequence ($K = 1, 2, 3$). With a fixed $d = 27.5m$ we get:

Speed (m/s)	$K = 1$	$K = 2$	$K = 3$
$i = 1$	69.3	34.65	23.1
$i = 2$	144.1	72.05	48.03

Table 7.1: Critical Velocities (m/s)

Speed (km/h)	$K = 1$	$K = 2$	$K = 3$
$i = 1$	249.5	124.7	83.16
$i = 2$	518.8	259.4	155.1

Table 7.2: Critical Velocities (km/h)

In order to **fix a speed limit** for the trains that are crossing the bridge we must be sure to be far enough to each of the value in the Table 7.2. With a speed **under 80 km/h** we are reasonably sure to not excite no one of the first two natural frequency of the bridge. We can extend the analysis for all the 6 natural frequency.

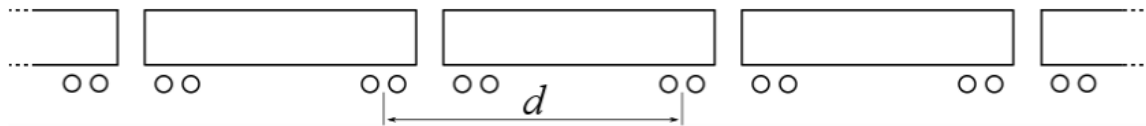


Figure 7.1: Spacing of the loads

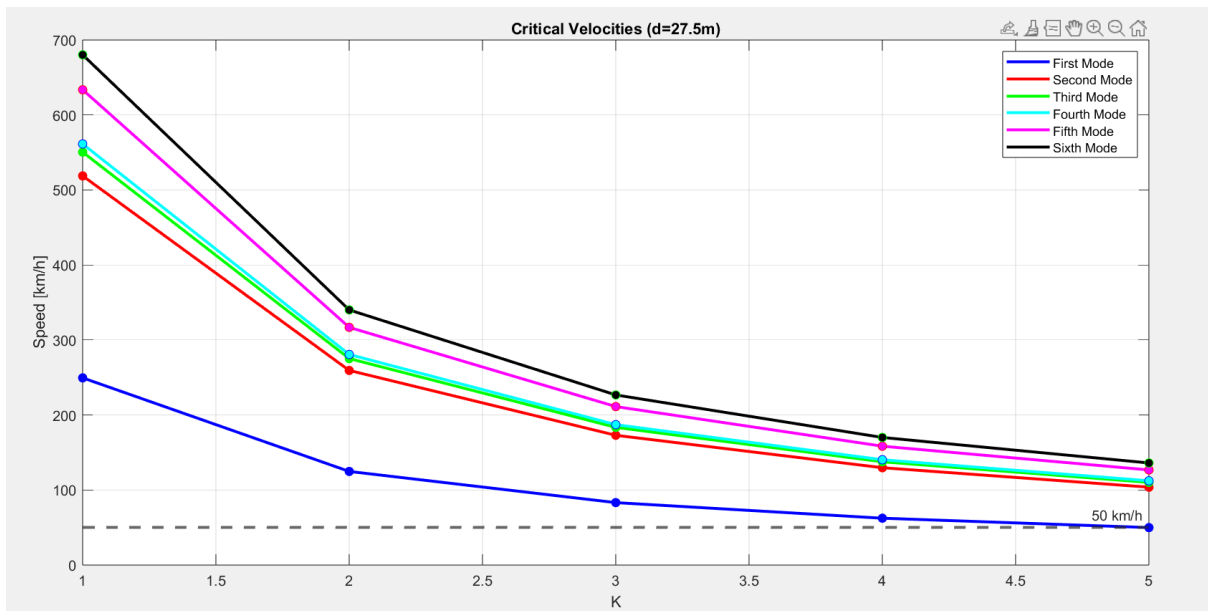


Figure 7.2: Critical velocities